Dictionaries and Sets

Chapter 06

6.1 Introduction

- A **dictionary** is an *unordered* collection which stores **key–value pairs** that map immutable keys to values, just as a conventional dictionary maps words to definitions.
- A **set** is an unordered collection of unique immutable elements.

6.2 Dictionaries

- A dictionary *associates* keys with values.
- Each key *maps* to a specific value.
- Sample dictionary keys and values:

Keys	Key type	Values	Value type
Country names	`str`	Internet country codes	`str`
Decimal numbers	`int`	Roman numerals	`str`
States	`str`	Agricultural products	list of `str`
Hospital patients	`str`	Vital signs	tuple of `int` s and `float` s
Baseball players	`str`	Batting averages	`float`
Metric measurements	`str`	Abbreviations	`str`
Inventory codes	`str`	Quantity in stock	`int`

Unique Keys

- Keys must be *immutable* and *unique*.
- Multiple keys can have the same value.

6.2.1 Creating a Dictionary

- Create a dictionary by enclosing in curly braces, `{}`, a comma-separated list of key–value pairs, each of the form *key*: *value*.
- Create an empty dictionary with `{}`.

```
In [1]: country_codes = {'Finland': 'fi', 'South Africa': 'za',
                         'Nepal': 'np'}
```

```
In [2]: country_codes
```

```
Out[2]: {'Finland': 'fi', 'South Africa': 'za', 'Nepal': 'np'}
```

- Dictionaries are *unordered* collections.
- Do *not* write code that depends on the order of the key–value pairs.

Determining if a Dictionary Is Empty

In [3]:
```
len(country_codes)
```

Out[3]:
```
3
```

- Can use a dictionary as a condition to determine if it's empty—non-empty is `True` and empty is `False`

In [4]:
```
if country_codes:
    print('country_codes is not empty')
else:
    print('country_codes is empty')
```

```
country_codes is not empty
```

In [5]:
```
country_codes.clear()
```

In [6]:
```
if country_codes:
    print('country_codes is not empty')
else:
    print('country_codes is empty')
```

```
country_codes is empty
```

6.2.2 Iterating through a Dictionary

In [1]:
```
days_per_month = {'January': 31, 'February': 28, 'March': 31}
```

In [2]:
```
days_per_month
```

Out[2]:
```
{'January': 31, 'February': 28, 'March': 31}
```

- Dictionary method `items` returns each key–value pair as a tuple:

In [3]:
```
for month, days in days_per_month.items():
    print(f'{month} has {days} days')
```

```
January has 31 days
February has 28 days
March has 31 days
```

6.2.3 Basic Dictionary Operations

- Intentionally provided the incorrect value `100` for the key `'X'`:

```
In [1]: roman_numerals = {'I': 1, 'II': 2, 'III': 3, 'V': 5, 'X': 100}

In [2]: roman_numerals
Out[2]: {'I': 1, 'II': 2, 'III': 3, 'V': 5, 'X': 100}
```

Accessing the Value Associated with a Key

```
In [3]: roman_numerals['V']
Out[3]: 5
```

Updating the Value of an Existing Key–Value Pair

```
In [4]: roman_numerals['X'] = 10

In [5]: roman_numerals
Out[5]: {'I': 1, 'II': 2, 'III': 3, 'V': 5, 'X': 10}
```

Adding a New Key–Value Pair

```
In [6]: roman_numerals['L'] = 50

In [7]: roman_numerals
Out[7]: {'I': 1, 'II': 2, 'III': 3, 'V': 5, 'X': 10, 'L': 50}
```

- String keys are case sensitive.
- Assigning to a nonexistent key inserts a new key–value pair.

Removing a Key–Value Pair

```
In [8]: del roman_numerals['III']

In [9]: roman_numerals
Out[9]: {'I': 1, 'II': 2, 'V': 5, 'X': 10, 'L': 50}
```

- Method **`pop`** returns the value for the removed key.

```
In [10]: roman_numerals.pop('X')
Out[10]: 10

In [11]: roman_numerals
Out[11]: {'I': 1, 'II': 2, 'V': 5, 'L': 50}
```

Attempting to Access a Nonexistent Key

```
In [12]: roman_numerals['III']
```

```
---------------------------------------------------------------------------
KeyError                                  Traceback (most recent call last)
<ipython-input-12-ccd50c7f0c8b> in <module>
----> 1 roman_numerals['III']

KeyError: 'III'
```

- Method **`get`** returns its argument's corresponding value or `None` if the key is not found.
- IPython does not display anything for `None`.
- `get` with a second argument returns teh second argument if the key is not found.

```
In [13]: roman_numerals.get('III')
```

```
In [14]: roman_numerals.get('III', 'III not in dictionary')
Out[14]: 'III not in dictionary'
```

```
In [15]: roman_numerals.get('V')
Out[15]: 5
```

Testing Whether a Dictionary Contains a Specified Key

```
In [16]: 'V' in roman_numerals
Out[16]: True

In [17]: 'III' in roman_numerals
Out[17]: False

In [18]: 'III' not in roman_numerals
Out[18]: True
```

6.2.4 Dictionary Methods `keys` and `values`

In [1]:
```
months = {'January': 1, 'February': 2, 'March': 3}
```

In [2]:
```
for month_name in months.keys():
    print(month_name, end='  ')
```

```
January  February  March
```

In [3]:
```
for month_number in months.values():
    print(month_number, end='  ')
```

```
1  2  3
```

Dictionary Views

- Methods `items`, `keys` and `values` each return a **view** of a dictionary's data.
- When you iterate over a **`view`**, it "sees" the dictionary's **current contents**—it does **not** have its own copy of the data.

In [4]:
```
months_view = months.keys()
```

In [5]:
```
for key in months_view:
    print(key, end='  ')
```

```
January  February  March
```

In [6]:
```
months['December'] = 12
```

In [7]:
```
months
```

Out[7]:
```
{'January': 1, 'February': 2, 'March': 3, 'December': 12}
```

In [8]:
```
for key in months_view:
    print(key, end='  ')
```

```
January  February  March  December
```

Converting Dictionary Keys, Values and Key–Value Pairs to Lists

```
In [9]: list(months.keys())
Out[9]: ['January', 'February', 'March', 'December']

In [10]: list(months.values())
Out[10]: [1, 2, 3, 12]

In [11]: list(months.items())
Out[11]: [('January', 1), ('February', 2), ('March', 3), ('December', 12)]
```

Processing Keys in Sorted Order

```
In [12]: for month_name in sorted(months.keys()):
             print(month_name, end='  ')
December  February  January  March
```

6.2.5 Dictionary Comparisons

- `==` is `True` if both dictionaries have the same key–value pairs, ***regardless* of the order in which those key–value pairs were added to each dictionary**.

In [1]:
```
country_capitals1 = {'Belgium': 'Brussels',
                     'Haiti': 'Port-au-Prince'}
```

In [2]:
```
country_capitals2 = {'Nepal': 'Kathmandu',
                     'Uruguay': 'Montevideo'}
```

In [3]:
```
country_capitals3 = {'Haiti': 'Port-au-Prince',
                     'Belgium': 'Brussels'}
```

In [4]:
```
country_capitals1 == country_capitals2
```

Out[4]: False

In [5]:
```
country_capitals1 == country_capitals3
```

Out[5]: True

In [6]:
```
country_capitals1 != country_capitals2
```

Out[6]: True

6.2.6 Example: Dictionary of Student Grades

- Script with a dictionary that represents an instructor's grade book.
- Maps each student's name (a string) to a list of integers containing that student's grades on three exams.

```python
# fig06_01.py
"""Using a dictionary to represent an instructor's grade book."""
grade_book = {
    'Susan': [92, 85, 100],
    'Eduardo': [83, 95, 79],
    'Azizi': [91, 89, 82],
    'Pantipa': [97, 91, 92]
}

all_grades_total = 0
all_grades_count = 0

for name, grades in grade_book.items():
    total = sum(grades)
    print(f'Average for {name} is {total/len(grades):.2f}')
    all_grades_total += total
    all_grades_count += len(grades)

print(f"Class's average is: {all_grades_total / all_grades_count:.2f}")
```

```
In [66]: run fig06_01.py

Average for Susan is 92.33
Average for Eduardo is 85.67
Average for Azizi is 87.33
Average for Pantipa is 93.33
Class's average is: 89.67
```

6.2.7 Example: Word Counts

- Script that builds a dictionary to count the number of occurrences of each word in a **tokenized** string.
- Python automatically concatenates strings separated by whitespace in parentheses.

```
# fig06_02.py
"""Tokenizing a string and counting unique words."""

text = ('this is sample text with several words '
        'this is more sample text with some different words')

word_counts = {}

# count occurrences of each unique word
for word in text.split():
    if word in word_counts:
        word_counts[word] += 1  # update existing key-value pair
    else:
        word_counts[word] = 1  # insert new key-value pair

print(f'{"WORD":<12}COUNT')

for word, count in sorted(word_counts.items()):
    print(f'{word:<12}{count}')

print('\nNumber of unique words:', len(word_counts))
```

```
In [1]: run fig06_02.py
WORD        COUNT
different   1
is          2
more        1
sample      2
several     1
some        1
text        2
this        2
with        2
words       2

Number of unique words: 10
```

Python Standard Library Module `collections`

- The Python Standard Library already contains the counting functionality shown above.
- A **`Counter`** is a customized dictionary that receives an iterable and summarizes its elements.

In [2]:
```
from collections import Counter
```

In [3]:
```
text = ('this is sample text with several words '
        'this is more sample text with some different words')
```

In [4]:
```
counter = Counter(text.split())
```

In [5]:
```
for word, count in sorted(counter.items()):
    print(f'{word:<12}{count}')
```

```
different   1
is          2
more        1
sample      2
several     1
some        1
text        2
this        2
with        2
words       2
```

In [6]:
```
print('Number of unique keys:', len(counter.keys()))
```

```
Number of unique keys: 10
```

6.2.8 Dictionary Method `update`

- Can insert and update key–value pairs.
- Method `update` also can receive an iterable object containing key–value pairs, such as a list of two-element tuples.

```
In [1]: country_codes = {}
```

```
In [2]: country_codes.update({'South Africa': 'za'})
```

```
In [3]: country_codes
```

```
Out[3]: {'South Africa': 'za'}
```

- Purposely inserting an incorrect value:

```
In [4]: country_codes.update(Australia='ar')
```

```
In [5]: country_codes
```

```
Out[5]: {'South Africa': 'za', 'Australia': 'ar'}
```

- Correcting the incorrect value:

```
In [6]: country_codes.update(Australia='au')
```

```
In [7]: country_codes
```

```
Out[7]: {'South Africa': 'za', 'Australia': 'au'}
```

6.2.9 Dictionary Comprehensions

- Convenient notation for quickly generating dictionaries, often by **mapping** one dictionary to another.
- The expression to the left of the `for` clause specifies a **key–value pair of the form *key* `:` *value***.
- In a dictionary with ***unique* values**, you can **reverse** the key–value pair mappings.

```
In [1]: months = {'January': 1, 'February': 2, 'March': 3}
```

```
In [2]: months2 = {number: name for name, number in months.items()}
```

```
In [3]: months2
```

```
Out[3]: {1: 'January', 2: 'February', 3: 'March'}
```

- A dictionary comprehension also can map a dictionary's values to new values.

```
In [4]: grades = {'Sue': [98, 87, 94], 'Bob': [84, 95, 91]}
```

```
In [5]: grades2 = {k: sum(v) / len(v) for k, v in grades.items()}
```

```
In [6]: grades2
```

```
Out[6]: {'Sue': 93.0, 'Bob': 90.0}
```

6.3 Sets

- A set is an unordered collection of **unique values**.
- Sets do not support indexing and slicing.

Creating a Set with Curly Braces

- Duplicates are ignored, making sets great for **duplicate elimination**.

```
In [1]: colors = {'red', 'orange', 'yellow', 'green', 'red', 'blue'}
```

- Though the output below is sorted, sets are **unordered**—do not write order-dependent code.

```
In [2]: colors
Out[2]: {'blue', 'green', 'orange', 'red', 'yellow'}
```

Determining a Set's Length

```
In [3]: len(colors)
Out[3]: 5
```

Checking Whether a Value Is in a Set

```
In [4]: 'red' in colors
```

```
Out[4]: True
```

```
In [5]: 'purple' in colors
```

```
Out[5]: False
```

```
In [6]: 'purple' not in colors
```

```
Out[6]: True
```

Iterating Through a Set

- There's no significance to the iteration order.

```
In [7]: for color in colors:
            print(color.upper(), end=' ')
```

```
YELLOW BLUE GREEN RED ORANGE
```

Creating a Set with the Built-In `set` Function

```
In [8]: numbers = list(range(10)) + list(range(5))
```

```
In [9]: numbers
```

```
Out[9]: [0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 0, 1, 2, 3, 4]
```

```
In [10]: set(numbers)
```

```
Out[10]: {0, 1, 2, 3, 4, 5, 6, 7, 8, 9}
```

- To create an empty set, must use the **`set()`**, because **`{}` represents an empty dictionary**.
- Python displays an empty set as `set()` to avoid confusion with an empty dictionary (`{}`).

```
In [11]: set()
```

```
Out[11]: set()
```

6.3.2 Mathematical Set Operations

Union

- The **union** of two sets is a set consisting of all the unique elements from both sets.
- The union operator requires two sets, but method `union` may receive any iterable as its argument (this is true for subsequent methods in this section as well).

```
In [1]: {1, 3, 5} | {2, 3, 4}
Out[1]: {1, 2, 3, 4, 5}

In [2]: {1, 3, 5}.union([20, 20, 3, 40, 40])
Out[2]: {1, 3, 5, 20, 40}
```

Intersection

The **intersection** of two sets is a set consisting of all the unique elements that the two sets have in common.

```
In [3]: {1, 3, 5} & {2, 3, 4}
Out[3]: {3}

In [4]: {1, 3, 5}.intersection([1, 2, 2, 3, 3, 4, 4])
Out[4]: {1, 3}
```

Difference

The **difference** between two sets is a set consisting of the elements in the left operand that are not in the right operand.

```
In [5]: {1, 3, 5} - {2, 3, 4}
Out[5]: {1, 5}

In [6]: {1, 3, 5, 7}.difference([2, 2, 3, 3, 4, 4])
Out[6]: {1, 5, 7}
```

6.3.3 Mutable Set Operators and Methods

Mutable Mathematical Set Operations

```
In [1]: numbers = {1, 3, 5}
```

```
In [2]: numbers |= {2, 3, 4}
```

```
In [3]: numbers
```

```
Out[3]: {1, 2, 3, 4, 5}
```

```
In [4]: numbers.update(range(10))
```

```
In [5]: numbers
```

```
Out[5]: {0, 1, 2, 3, 4, 5, 6, 7, 8, 9}
```

Methods for Adding and Removing Elements

- Set method **`add`** inserts its argument if the argument is *not* already in the set; otherwise, the set remains unchanged.

```
In [6]: numbers.add(17)

In [7]: numbers.add(3)

In [8]: numbers
Out[8]: {0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 17}
```

- Method **`remove`** removes its argument from the set—raises a **`KeyError`** if the value is not in the set.

```
In [9]: numbers.remove(3)

In [10]: numbers
Out[10]: {0, 1, 2, 4, 5, 6, 7, 8, 9, 17}
```

- Method `discard` also removes its argument from the set but **does not cause an exception if the value is not in the set**.
- Can remove an *arbitrary* set element and return it with `pop`.

```
In [11]: numbers.pop()
Out[11]: 0

In [12]: numbers
Out[12]: {1, 2, 4, 5, 6, 7, 8, 9, 17}

In [13]: numbers.clear()  # empty the set

In [14]: numbers
Out[14]: set()
```

6.3.4 Set Comprehensions

- Define set comprehensions in curly braces.

In [1]:
```
numbers = [1, 2, 2, 3, 4, 5, 6, 6, 7, 8, 9, 10, 10]
```

In [2]:
```
evens = {item for item in numbers if item % 2 == 0}
```

In [3]:
```
evens
```

Out[3]:
```
{2, 4, 6, 8, 10}
```

6.4 Intro to Data Science: Dynamic Visualizations

- In this section, we make things “come alive” with *dynamic visualizations*.

The Law of Large Numbers

- For a six-sided die, each value 1 through 6 should occur one-sixth of the time, so the probability of any one of these values occurring is 1/6th or about 16.667%.
- In the dynamic visualization, the more rolls we perform, the closer each die value’s percentage of the total rolls gets to 16.667% and the heights of the bars gradually become about the same.
- This is a manifestation of the *law of large numbers*.

6.4.1 How Dynamic Visualization Works

- The Matplotlib `animation` module’s `FuncAnimation` function, updates a visualization *dynamically*.

Animation Frames

- `FuncAnimation` drives a **frame-by-frame animation**.
- Each **animation frame** specifies what to change during one plot update.
- Stringing together many updates over time creates an animation.
- This example displays an animation frame every 33 milliseconds—yielding approximately 30 (1000 / 33) frames-per-second.

Running `RollDieDynamic.py`

1. Access the command line in Jupyter with **File > New > Terminal**.
2. `cd ch06`.
3. Execute

```
ipython RollDieDynamic.py 6000 1
```

- 6000 is the number of animation frames to display.
- 1 is the number of die rolls to summarize in each animation frame.

- To see the law of large numbers in action, increase the execution speed by rolling the die more times per animation frame:

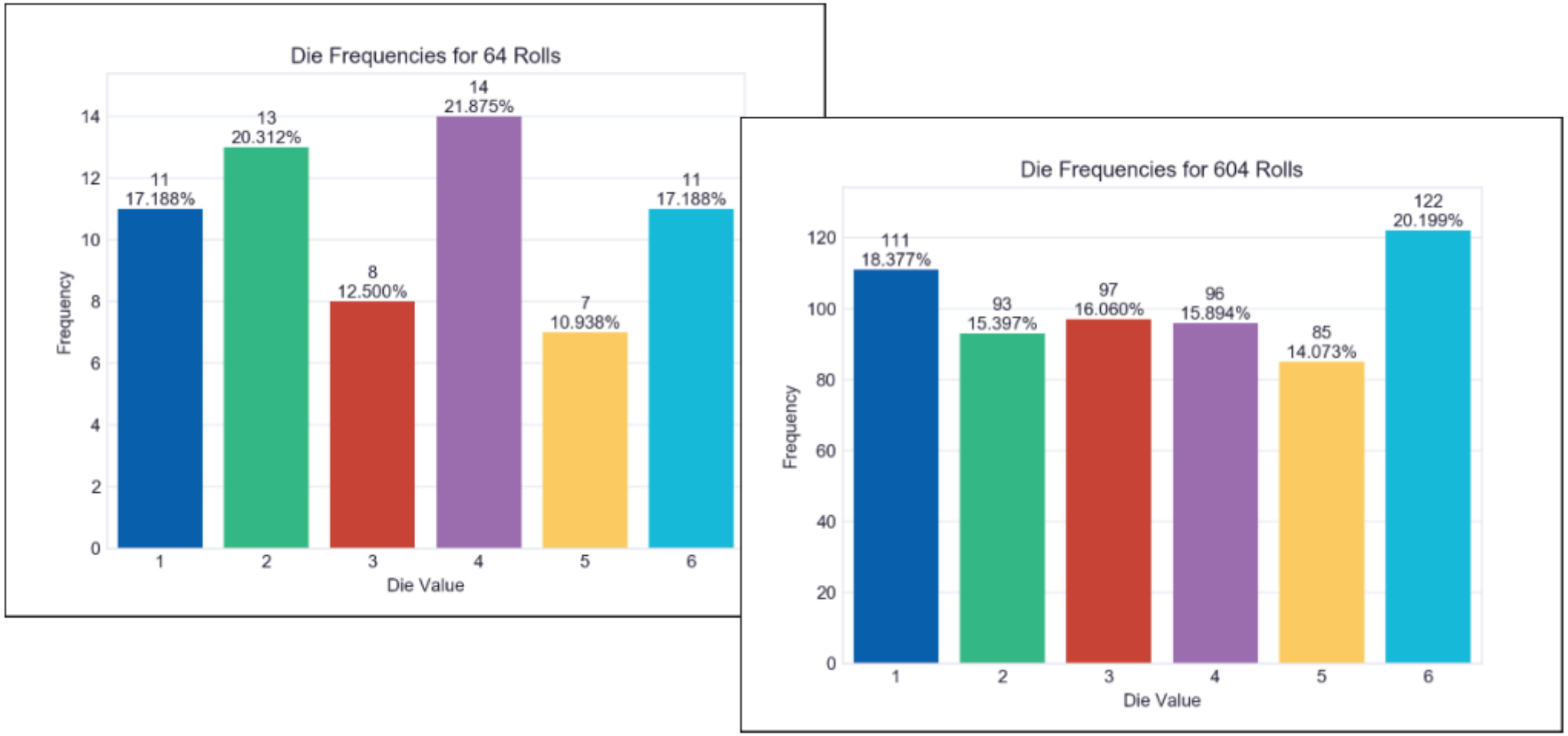
```
ipython RollDieDynamic.py 10000 600
```

- In this case, `FuncAnimation` perform 10,000 updates, with 600 rolls-per-frame for a total of 6,000,000 rolls.

Sample Executions

Execute 6000 animation frames rolling the die once per frame:

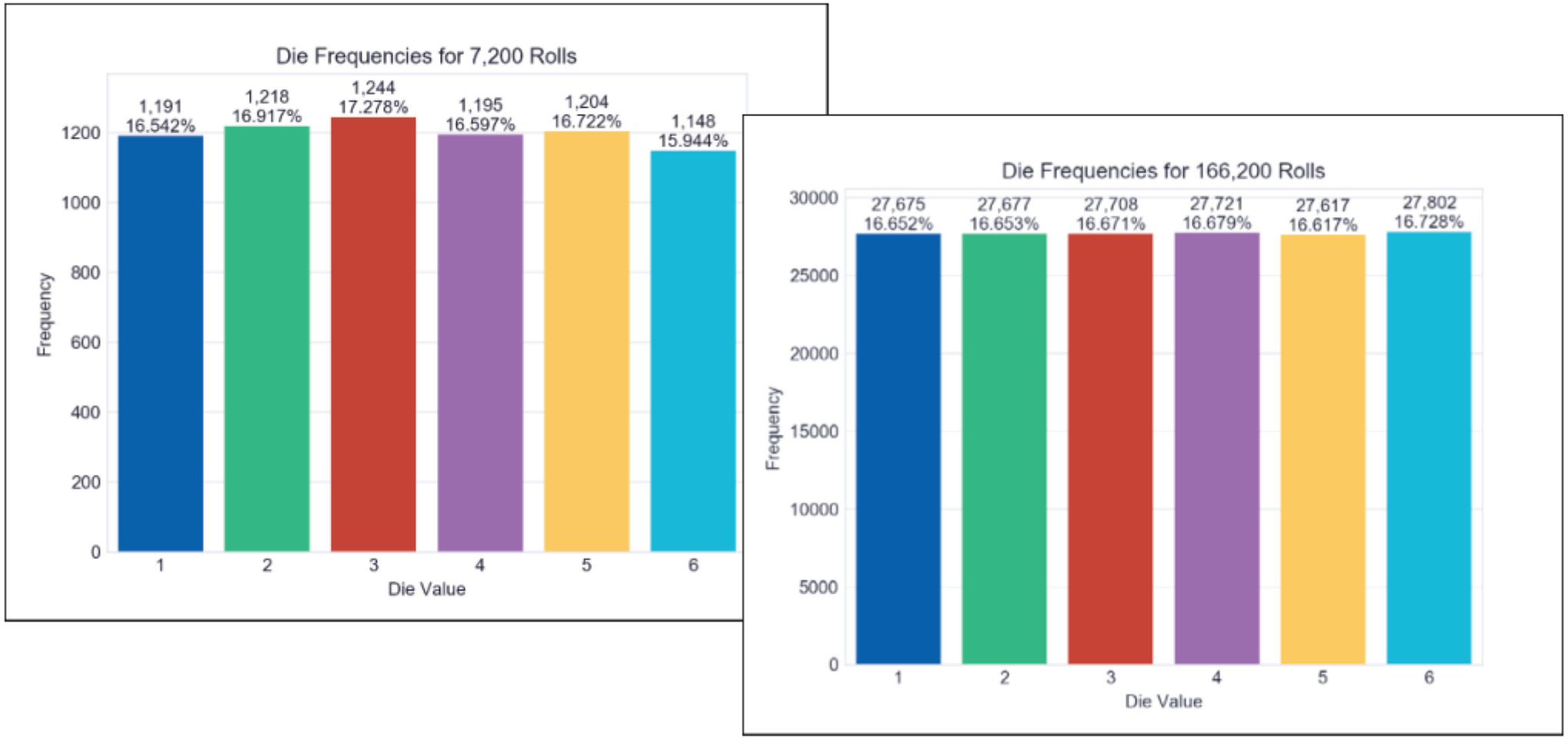
```
ipython RollDieDynamic.py 6000 1
```



Sample Executions

Execute 10,000 animation frames rolling the die 600 times per frame:

```
ipython RollDieDynamic.py 10000 600
```



6.4.2 Implementing a Dynamic Visualization

Importing the Matplotlib `animation` Module

- We focus primarily on the new features used in this example.
- We import the Matplotlib `animation` module to access `` `FuncAnimation ``.

```
# RollDieDynamic.py
"""Dynamically graphing frequencies of die rolls."""
from matplotlib import animation
import matplotlib.pyplot as plt
import random
import seaborn as sns
import sys
```

Function `update`

```
    def update(frame_number, rolls, faces, frequencies):
        """Configures bar plot contents for each animation frame."""
```

- `FuncAnimation` calls the `update` function once per animation frame.
- This function must receive at least one argument.
- Parameters:
  - `frame_number`—The next value from `FuncAnimation`'s `frames` argument, which we'll discuss momentarily. Though `FuncAnimation` requires the `update` function to have this parameter, we do not use it in this `update` function.
  - `rolls`—The number of die rolls per animation frame.
  - `faces`—The die face values used as labels along the graph's *x*-axis.
  - `frequencies`—The list in which we summarize the die frequencies.

We discuss the rest of the function's body below.

Function `update`: Rolling the Die and Updating the `frequencies` List

- Roll the die `rolls` times and increment the appropriate `frequencies` element for each roll.

```
    # roll die and update frequencies
        for i in range(rolls):
            frequencies[random.randrange(1, 7) - 1] += 1
```

Function `update`: Configuring the Bar Plot and Text

- The `matplotlib.pyplot` module's **`cla`** (clear axes) function removes the existing bar plot elements before drawing new ones for the current animation frame.
- We discussed the other code below in the previous chapter's Intro to Data Science section.

```
# reconfigure plot for updated die frequencies
    plt.cla()  # clear old contents contents of current Figure
    axes = sns.barplot(faces, frequencies, palette='bright')  # new bars
    axes.set_title(f'Die Frequencies for {sum(frequencies):,} Rolls')
    axes.set(xlabel='Die Value', ylabel='Frequency')
    axes.set_ylim(top=max(frequencies) * 1.10)  # scale y-axis by 10%

    # display frequency & percentage above each patch (bar)
    for bar, frequency in zip(axes.patches, frequencies):
        text_x = bar.get_x() + bar.get_width() / 2.0
        text_y = bar.get_height()
        text = f'{frequency:,}\n{frequency / sum(frequencies):.3%}'
        axes.text(text_x, text_y, text, ha='center', va='bottom')
```

Variables Used to Configure the Graph and Maintain State

- The `sys` module's `argv` list contains the script's command-line arguments.
- The `matplotlib.pyplot` module's **`figure`** function gets a `Figure` object in which `FuncAnimation` displays the animation — one of `FuncAnimation`'s required arguments.

```
# read command-line arguments for number of frames and rolls per frame
number_of_frames = int(sys.argv[1])
rolls_per_frame = int(sys.argv[2])

sns.set_style('whitegrid')  # white background with gray grid lines
figure = plt.figure('Rolling a Six-Sided Die')  # Figure for animation
values = list(range(1, 7))  # die faces for display on x-axis
frequencies = [0] * 6  # six-element list of die frequencies
```

Calling the `animation` Module's `FuncAnimation` Function

- `FuncAnimation` returns an object representing the animation.
- Though this is not used explicitly, you *must* store the reference to the animation; otherwise, Python immediately terminates the animation and returns its memory to the system.

```
# configure and start animation that calls function update
die_animation = animation.FuncAnimation(
    figure, update, repeat=False, frames=number_of_frames, interval=33,
    fargs=(rolls_per_frame, values, frequencies))

plt.show()  # display window
```

`FuncAnimation` has two required arguments:

- `figure`—the `Figure` object in which to display the animation, and
- `update`—the function to call once per animation frame.
- Optional keyword arguments:
  - **`repeat`**—If `True` (the default), when the animation completes it restarts from the beginning.
  - **`frames`**—The total number of animation frames, which controls how many times `FunctAnimation` calls `update`.
  - **`interval`**—The number of milliseconds between animation frames (the default is 200).
  - **`fargs`** (short for "function arguments")—A tuple of other arguments to pass to the function you specified in `FuncAnimation`'s second argument.

```
# RollDieDynamic.py
"""Dynamically graphing frequencies of die rolls."""
from matplotlib import animation
import matplotlib.pyplot as plt
import random
import seaborn as sns
import sys

def update(frame_number, rolls, faces, frequencies):
    """Configures bar plot contents for each animation frame."""
    # roll die and update frequencies
    for i in range(rolls):
        frequencies[random.randrange(1, 7) - 1] += 1

    # reconfigure plot for updated die frequencies
    plt.cla()  # clear old contents contents of current Figure
    axes = sns.barplot(x=faces, y=frequencies, palette='bright')  # new bars
    axes.set_title(f'Die Frequencies for {sum(frequencies):,} Rolls')
    axes.set(xlabel='Die Value', ylabel='Frequency')
    axes.set_ylim(top=max(frequencies) * 1.10)  # scale y-axis by 10%

    # display frequency & percentage above each patch (bar)
    for bar, frequency in zip(axes.patches, frequencies):
        text_x = bar.get_x() + bar.get_width() / 2.0
        text_y = bar.get_height()
        text = f'{frequency:,}\n{frequency / sum(frequencies):.3%}'
        axes.text(text_x, text_y, text, ha='center', va='bottom')

# read command-line arguments for number of frames and rolls per frame
number_of_frames = int(sys.argv[1])
rolls_per_frame = int(sys.argv[2])

sns.set_style('whitegrid')  # white background with gray grid lines
figure = plt.figure('Rolling a Six-Sided Die')  # Figure for animation
values = list(range(1, 7))  # die faces for display on x-axis
frequencies = [0] * 6  # six-element list of die frequencies

# configure and start animation that calls function update
die_animation = animation.FuncAnimation(
    figure, update, repeat=False, frames=number_of_frames, interval=33,
    fargs=(rolls_per_frame, values, frequencies))

plt.show()  # display window
```